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Reinforcement learning-based drone-client selection for efficient federated learning-based intrusion detection in FANETs

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Abstract
Integrating Flying Ad hoc Network (FANET) technology with smart sensors and devices has sparked numerous innovative applications. However, this integration has exposed FANETs to significant security threats, undermining network integrity and reliability. Deep learning (DL) has recently garnered attention for intrusion detection systems (IDSs) within FANETs. While centralized DL (CDL) offers robustness and effectiveness, it raises concerns about privacy, computational overhead, and scalability. In contrast, federated deep learning (FDL) ensures data privacy and conserves drone energy, but faces deployment challenges in dynamic and resource-constrained FANETs, including communication constraints and drone heterogeneity. To address these challenges, optimal drone clients must be selected for participation in FDL, necessitating precise drone positioning. This paper proposes a reinforcement learning-based drone client selection algorithm to determine optimal drone-client participation in FDL training. Building upon this algorithm, we introduce a novel FDL-based IDS (FLID) for efficient detection of malicious attacks in FANETs with minimal computational overhead and data-privacy preservation, addressing communication and drone heterogeneity constraints. FLID incorporates multilayer perceptron, convolutional neural networks, and recurrent neural networks with long short-term memory using the WSN-BFSF dataset tailored for detecting flooding, blackhole, and selective forwarding attacks. Experimental results demonstrate that FLID fulfills IDS requirements and achieves superior performance compared to other reference models, with an accuracy of 99.38%, precision of 99%, recall of 99%, and an F1-score of 99%.

Keywords Deep learning (DL) · Drones · Federated DL (FDL) · Intrusion detection (ID) · Reinforcement learning (RL).

1 Introduction

Recently, significant advances have been made in robotics, particularly in drones. Drones are beneficial in various applications, including border patrol missions, target detection [1, 2], medical supplies delivery, and search and rescue operations [3, 4]. Flying ad hoc networks (FANETs) are custom ad hoc networks explicitly designed for drones [5]. However, using FANET and advanced communication technologies introduces several cybersecurity risks and weaknesses. FANETs become vulnerable to various attacks due to malicious devices in the network [6]. Some common attacks are jamming, flooding, selective forwarding, blackhole, wormholes, and Sybil attacks, which can cause severe performance degradation [7, 8].

A crucial security concern revolves around the vulnerability of FANETs to potential intrusions, which pose significant risks to public safety and data integrity. Given

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the growing need for drones in critical applications, ensuring the security and privacy of their operations is of utmost importance [9, 10]. Several researchers have conducted studies to improve FANET security and privacy preservation, explicitly focusing on intrusion detection systems (IDSs). IDS plays a crucial role in ensuring the security of FANETs. Researchers have proposed using supervised machine learning (ML) [11, 12] and deep learning (DL) [13–15] to design intelligent and accurate IDSs within FANETs. The application of DL can improve the self-learning capabilities of IDS and foster the development of more intelligent and adaptive security solutions to detect attacks within the network [16]. There are two prominent paradigms in deep learning: centralized deep learning (CDL) and federated deep learning (FDL) [17].

Centralized DL models train the model throughout the dataset, resulting in high accuracy, precision, recall, and F1-score. However, CDL does not meet the rigorous privacy constraints inherent in FANET systems, especially when dealing with sensitive information. Furthermore, the high processing requirements of these models and communication challenges resulting from moving large amounts of data to a central server can be costly in terms of time and resources, offering a considerable obstacle to IDS. As the network grows, the scalability of CDL-based IDS becomes a bottleneck, limiting its utility in large-scale FANET installations. To address these limitations, FDL has emerged as a decentralized ML paradigm that enables the development of more robust and accurate IDSs for FANETs. FDL offers the potential to improve privacy by enabling model training in a decentralized manner without sharing raw data [18]. Second, FDL reduces computational overhead, as only model updates are transmitted between the central server and local devices, thus minimizing the amount of data exchanged. Third, FDL does not affect scalability.

FDL has recently been proposed for attack detection within FANETs [19, 20], but may face two main limitations due to the unique characteristics of FANETs. The first limitation is related to low drone density with longer distances between them, which can lead to communication issues when FDL parameters are transmitted. Therefore, the position of each drone becomes crucial in selecting which drones to participate in the FDL process. The second limitation is related to drone heterogeneity; drones involved in the FDL process may exhibit heterogeneity in their capabilities, which vary over time. Drones may have different levels of computing power, constrained by limited battery and storage capacity. This heterogeneity requires an intelligent selection mechanism to determine which drones are suitable for participation.

Traditional static model-based selection methods struggle to effectively model the dynamics of drone networks, where devices face computational, storage, and energy constraints. However, determining the optimal decisions is challenging due to the stochastic nature of the FANET environment, where drone resource information is uncertain. Moreover, existing methods often rely on static algorithms, and intelligent algorithms were rarely applied in this context. To solve these limitations, the Multi-Access Edge Computing (MEC) server should strategically select drones that possess sufficient energy and storage before the beginning of the FDL process. In this paper, we develop a reinforcement learning (RL) framework that enables ground control stations (MEC servers) to learn drone behavior and make intelligent drone-client selections based on location, power capacity, and storage availability. We propose leveraging Q-learning to enable the MEC server to develop an optimal client selection policy within the FANET system without prior knowledge of network dynamics. Q-learning is a straightforward and effective RL algorithm that is relatively easy to implement and execute. Compared to other RL algorithms, it does not require complex computations or extensive parameter tuning. Q-learning may have lower memory requirements, whereas Q-values can be stored efficiently in a table. Specifically, we model the dynamics of the FANET system as a Markov Decision Process (MDP) and then apply Q-Learning with an ϵ-greedy policy to balance exploration and exploitation, allowing MEC servers to make appropriate decisions when selecting suitable drones for the FDL process. Motivated by these challenges, the main contributions of this paper can be summarized as follows:

- Developing an FDL approach for intrusion detection in FANETs named (FLID), using three types of deep neural networks: multilayer perceptron (MLP), recurrent neural network with long short-term memory (RNN+LSTM), and convolutional neural network (CNN). The purpose of this integration is to systematically evaluate and determine the most suitable model for FANET environments. By evaluating each model's performance under identical experimental conditions, we determine the optimal architecture in terms of precision, recall, and F1 score.
- Modeling the FANET network as an MDP without prior knowledge about the operating environment.
- Proposing an RL-based approach that enables the MEC server to learn from drone behavior and make intelligent drone-client selections before executing the FDL process. FLID, with the RL drone-client selection algorithm, meets IDS requirements such as privacy protection, effectiveness, robustness, context awareness, and computational efficiency.
- Evaluating the proposed model in FANET using the WSN-BFSF datasets designed to detect three attack

types: flooding attacks, blackhole attacks, and selective forwarding attacks. The results show that FLID outperforms other reference models in terms of accuracy, precision, recall, and F1-score.

The remaining sections of the paper are structured as follows: Section II discusses the existing CDL and FDL-based IDSs within FANET. Section III presents the system model. Section IV describes the problem statement. Section V presents the proposed solution. Experimental results and performance comparison are presented in Section VI. Finally, Section VII concludes the paper and provides future directions.

2 Related work

Traditional IDS strategies address several design problems, including the lack of crucial evaluation data, the need for big data processing, and the expectation of automation [21]. ML has been used for IDS in FANET to intelligently solve these issues. This section briefly reviews some of the existing ML-based IDSs in FANETs, focusing on CDL and FDL.

CDL-based Intrusion detection: Many researchers have proposed centralized deep learning for IDS. For example, Rabie et al. [13] introduced a real-time DL data analytics system and analyzed threats to FANET IDS. They use RNNs to find anomalies. Abdulghani et al. [14] explored an innovative strategy to improve IDS within drone communication channels. They used LSTM with the Synthetic Minority Oversampling Technique (SMOTE). They integrated a binary classification approach to improve the overall effectiveness of IDS. In a parallel effort, Escorcia et al. [15] used a technique to identify and classify intrusions within the Internet of Drones (IODs) called the Sea Turtle Foraging technique with Hybrid Deep Learning-based IDS. They used Deep Belief Network (DBN) and the Sparrow Search Optimization method. Omar et al. [22] introduced a streamlined module for IDS and prevention systems (IDPS) adapted for drones to improve deep learning-based IDS and ensure context awareness. They used Deep Q-learning (DQN), the IDPS module, to equip drones with the ability to independently identify suspicious activities and take appropriate measures to protect the network.

Similarly, Tao et al. [23] introduced a DRL methodology to identify malicious attacks within drone networks. Initially, they outlined the structure of drone computing networks and explored potential applications. Subsequently, they illustrated a case study using DRL for IDS, which improved security services. To increase data integrity within deep learning-based IDSs, Heidari et al. [24] introduced a novel approach to enhance the performance of the IoD device network using a blockchain-based neural network. This model improves data integrity and storage, facilitating intelligent decision-making in various IoD devices. Sorkhpour et al. [25] introduced Auto-CIDS, an autonomous cybersecurity intrusion detection system aimed at reducing human involvement and enabling active learning through the use of past experiences. Using DRL in conjunction with the strengths of unsupervised algorithms, they trained DQN agents in a self-supervised manner using their historical data.

The previous techniques have been widely used for IDS in FANETs, achieving good effectiveness. However, these models do not meet the rigorous privacy constraints inherent in FANET systems. Furthermore, the high processing requirements of these models and communication challenges resulting from moving large amounts of data to a central server can be costly in terms of time and resources, offering a notable obstacle for performing efficient IDSs. As the network grows, the scalability of CDL-based IDS becomes a bottleneck, limiting its applicability in large-scale FANETs.

FDL-based schemes: To the best of our knowledge, little research has been done on applying FDL-based attack detection in FANETs. For example, Nishat et al. [26] developed an adaptable novel protection system against jammer attacks based on FDL. They created a model-free Q-learning process controlled by an on-device federated jamming detection system and an adaptive ϵ-greedy strategy. Similarly, Wang et al.[19] suggested SFAC, a secure FDL system. They provided a blockchain-based collaborative learning architecture for drones that enabled safe sharing of local model updates and verification of contributions without a central curator.

Ntizikira et al. [20] used advanced methodologies such as FDL, differential privacy, and secure multiparty computation in a related context. A key aspect of their approach involved incorporating a sophisticated convolutional neural network with a long- and short-term memory network (CNN-LSTM). Namakshenas et al. [27] proposed the IP2FL framework, an Interpretation-driven privacy-preserving federated learning (FL) approach specifically designed for industrial cyber-physical systems. This framework integrates Additive Homomorphic Encryption to ensure privacy while employing advanced feature selection techniques and Shapley Values (SV) to enhance explainability. Yazdinejad et al. [28] presented a robust privacy-preserving FDL model designed to resist model poisoning attacks without compromising accuracy. They incorporated an internal auditor to evaluate the similarity and distribution of encrypted gradients, effectively distinguishing between benign and malicious gradients. They used Additive Homomorphic Encryption to ensure data

confidentiality while optimizing computational and communication efficiency. Danyal et al. [29] proposed a quantum-based registration and authentication mechanism to ensure stringent client verification within the FDL framework, addressing the critical need for robust client validation in FL. Additionally, they integrated Additive Homomorphic Encryption into their approach, effectively safeguarding clients' model weights and offering a privacy-preserving solution that maintains computational efficiency for FDL participants. Wang et al. [30] proposed an FL-based anomaly detection strategy for industrial IoT (IIoT) to improve security and privacy by using deep reinforcement learning (DRL) while minimizing data exposure. This approach achieves high throughput, low latency, and improved detection accuracy in diverse IIoT scenarios. Similarly, in [31], a heterogeneous blockchain-based hierarchical trust evaluation strategy was introduced to strengthen security in 5 G-enabled intelligent transportation systems (5 G-ITS). This is accomplished using FDL for trust assessment, integrating hierarchical incentive mechanisms, and ensuring trust verification through blockchain, resulting in high system throughput and low latency. The work in [32] introduced a blockchain-based secure data aggregation strategy (BSDA) for edge computing-enabled IoT, addressing privacy by limiting task receivers with security labels and creating new block generation rules. BSDA uses an improved deep deterministic policy gradient for energy-efficient routing, achieving high throughput, low latency, and excellent aggregation and energy efficiency. On the other hand, several FDL-based strategies have been proposed to address security challenges in Vehicular Ad hoc Networks (VANETs) [33, 34]. However, these methods are not directly applicable to FANETs due to two distinct characteristics. Unlike VANETs, FANETs have low node density and longer inter-node distances, posing communication reliability challenges. FANETs also prioritize energy efficiency to extend network lifetime, a concern not shared by VANETs. These differences highlight the need for FDL-based security solutions customized for FANETs.

In summary, FDL provides an effective, robust and privacy-preserving solution for FANET security, ensuring low computational overhead and scalability. However, optimizing FDL in dynamic and heterogeneous FANET environments remains a challenge. To address this, we propose an FDL-based IDS enhanced with an RL-driven drone-client selection algorithm, improving adaptability and communication efficiency. The model satisfies key IDS criteria, including accuracy, precision, recall, and F1-score, while effectively mitigating attacks such as flooding, blackholes, and selective forwarding. Using FDL, privacy is preserved, while RL enhances context awareness, maintaining efficiency without compromising scalability. Table 1 compares various existing FANET IDS models with our approach, emphasizing key trade-offs. This table indicates that CDL methods incur high computational costs, making them unsuitable for resource-constrained FANETs. In contrast, the table demonstrates that DRL enhances scalability but lacks privacy protection, while blockchain-enhanced CDL ensures privacy, yet remains computationally expensive. FL-based models maintain privacy but depend on static client selection, limiting adaptability. Our model addresses these challenges through RL-driven dynamic client selection, optimizing resource utilization, reducing computational overhead, and enhancing resilience against significant attacks. This guarantees an adaptive, efficient, and privacy-preserving IDS designed for dynamic FANET environments.

3 System model

A FANET with several drones operating in a mission area is considered. The mission area is divided into $n+1$ identical subareas, where $X \in \{x_0, x_1, x_2, ...x_n\}$. The mission area contains several randomly deployed sensor devices on the ground, and multiple drones fly in the sky to provide wireless services to them. We assume that each drone k is equipped with a limited rechargeable battery. This battery has a maximum capacity of $E_{\max}$. Furthermore, each drone has restricted storage availability, denoted by $M_{\max}$. The drone battery level is discretized into L positive levels $\{0, E^1, \ldots, E^L = E^{\max}\}$, where E^j denotes the jth stored energy level. Similarly, the storage capacity of each drone is discretized into Z levels $\{0, M^1, \ldots, M^Z = M^{\max}\}$, where M^j represents a positive integer indicating the jth level of storage availability. The battery level and memory space of drone k at time t, denoted by $E_k(t)$ and $M_k(t)$, respectively, can assume any of the available levels L and Z.

The FANET is controlled by ground MEC servers, which are crucial for managing and coordinating drone operations. The location of each drone k (or a MEC server) can be within any subregions (with coordinates (x, y, z)) at time slot t as:

$$\theta_k(t) = \{x_k, y_k, z_k\}. \tag{1}$$

Drones are specifically used to aggregate sensor data and transmit it to MEC ground servers. These devices bridge sensors and MEC servers and ensure that data are transmitted reliably and securely. We also consider cloud computing to allow the accessible collection and management of large amounts of data from MEC servers. These data can be accessed live from any location. The network model used in this paper is shown in Fig. 1. In our system

Table 1 Comparison between related works and the proposed model

Schemes	Scalability	Effectiveness				Robustness (Attack types)	Privacy protection	Context-awareness	Computational overhead
		Acc	Pre	Rec	Fls				
CDL (RNN) [13]	x	✓	✓	✓	✓	Intrusion attacks	x	x	High
CDL (LSTM) [14]	x	✓	✓	✓	✓	Intrusion attacks	x	x	High
CDL (DBN) [15]	x	✓	x	x	✓	Intrusion attacks	x	x	High
DRL [22]	x	✓	✓	✓	✓	Intrusion attacks	x	✓	High
DRL [23]	x	x	x	x	x	Intrusion attacks	x	✓	High
CDL-Blockchain [24]	x	✓	✓	✓	✓	DDoS, Botnet, Port scan	✓	x	High
FRL [26]	✓	✓	x	x	x	Jamming attack	✓	✓	Low
FDL [19]	✓	x	x	x	x	Privacy leakage	✓	x	Low
FDL (CNN+LSTM) [20]	✓	✓	✓	✓	✓	DDoS, MitM, Spoofing	✓	x	Low
The proposed model (RL+FDL)	✓	✓	✓	✓	✓	Flooding, Blackhole, Selective forwarding	✓	✓	Low

model, we consider the presence of a charging station (CH) within the mission area for drones to recharge. We also assume that MEC servers remain uncompromised, serving as FL aggregators for network security in the FANET, thus maintaining trustworthiness. Moreover, the FDL system incorporates security measures such as secure multiparty computation (SMC) [35] to facilitate the safe exchange of ML parameters from drones to the server. We consider the presence of malicious drones within the network, making FANET susceptible to various intrusion attacks and vulnerable to potential damage.

3.1 Attack models

We consider the presence of malicious drones within the network, making FANET susceptible to various intrusion attacks and vulnerable to potential damage. Three different types of attacks are considered: *flooding*, *blackhole*, and *selective forwarding* attacks.

3.1.1 Flooding attacks

Flooding attacks involve a malicious drone that floods the network with numerous route requests, leading to congestion and resource depletion between drones, compromising FANET availability and reliability, and increasing delay and packet loss [36].

3.1.2 Blackhole attacks

Blackhole attacks involve a malicious drone that falsely advertises itself as having the shortest or best route to the destination and drops all packets it receives from other drones. This attack can affect the integrity and confidentiality of FANET, reduce throughput, and increase packet loss [37].

3.1.3 Selective forwarding attacks

Selective forwarding attacks occur when a malicious drone selectively forwards certain packets while discarding others, resulting in network disruption, compromised integrity and reliability, poor quality of service, and increased packet loss [38].

3.2 Privacy attacks

Privacy attacks in FANET pose critical challenges due to the open and dynamic nature of the network. Adversaries can exploit wireless communication channels to perform eavesdropping, intercepting sensitive data exchanged between drones and ground control stations. Identity spoofing and Sybil attacks enable attackers to impersonate legitimate drones or flood the network with fake identities, disrupting communication and exposing private information. Furthermore, data manipulation and replay attacks compromise the integrity and confidentiality of transmitted data. Privacy protection in FANETs is crucial to safeguard sensitive data and ensure mission integrity. Techniques such as encryption and secure communication protocols form the foundational layer of privacy protection, ensuring that transmitted data remains confidential [28]. Advanced methods like FL enhance privacy by allowing decentralized model training, keeping raw data localized to drones, and reducing exposure to attacks [39]. Integrating differential privacy and secure aggregation with FL further

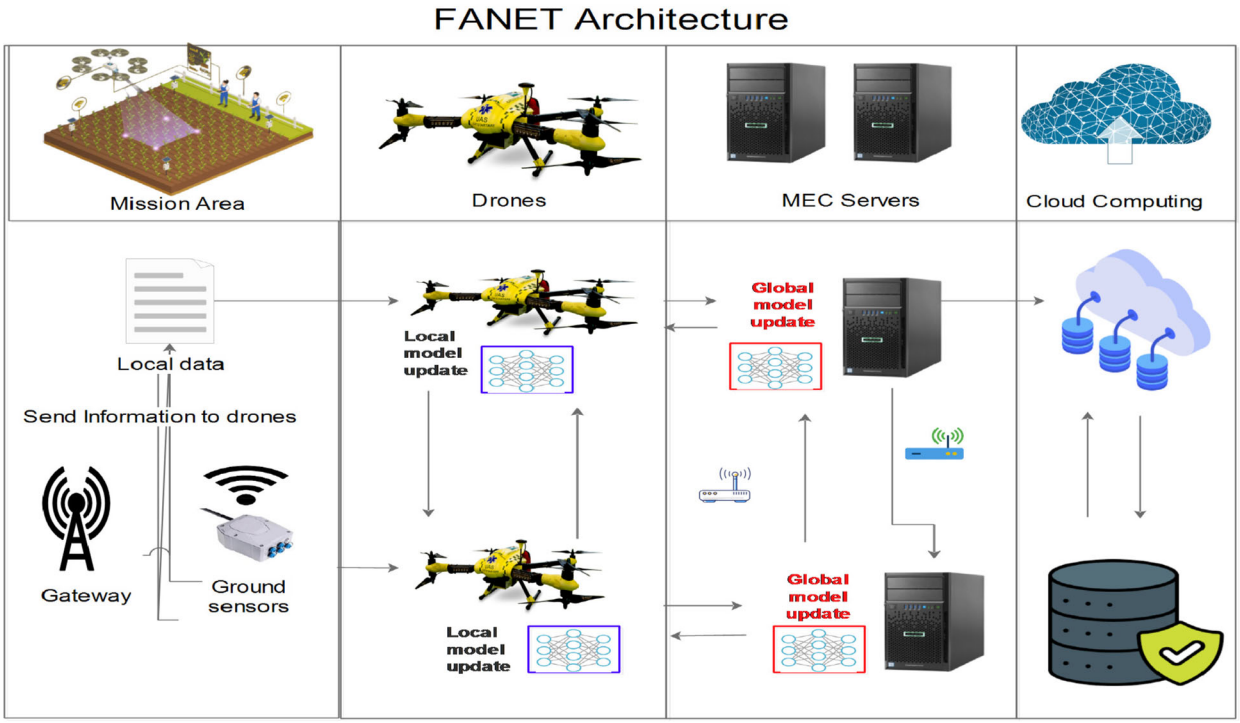


Fig. 1 Network model

protects against inference and reconstruction attacks [40]. Blockchain technology adds another layer of privacy and security by providing a decentralized, immutable ledger for data sharing and identity management, preventing tampering and unauthorized access[41].

4 Problem statement

The problem statement can be concisely stated as follows: In the context of a FANET that operates under various attack scenarios, the objective is to design an intelligent FDL-based IDS that is capable of efficiently detecting malicious attacks (i.e., intrusions within FANETs) with minimal computational overhead while preserving data privacy without affecting scalability. In addition, FDL-based IDS should address the following design challenges associated with applying FDL techniques in a dynamic FANET environment:

Communication challenge: FANETs are characterized by low node density, with longer distances between drones. This leads to communication issues between the server and drones while FDL parameters are being transmitted.

Resource heterogeneity challenge: The drones involved in the FDL process exhibit varying capabilities over time. Drones may have different levels of computing power, constrained by limited battery and storage capacity.

The challenge posed by communication constraints and drone heterogeneity requires an intelligent selection mechanism to identify drones suitable for participation in the FDL process. Consequently, the precise positioning of each drone is critical in determining its eligibility for participation in the FDL. To address communication and resource-constrained challenges within FANETs when applying FDL-based IDS, the MEC server should select the optimal drone client to participate in the FDL process.

5 The proposed solution

To address communication and resource-constrained challenges within FANETs when applying FDL-based IDS, we propose a Q-learning with a ϵ-greedy policy to allow the MEC server to select the optimal drone client to participate in the FDL process. The proposed FDL strategy classifies network traffic into four categories: normal, selective forwarding, blackhole, and flooding attack. We use three types of neural networks for anomaly detection: multilayer perceptron (MLP), convolutional neural networks(CNN), and recurrent neural networks with long short-term memory (RNN+LSTM) to capture temporal and spatial features. We evaluate the proposed method on a publicly available dataset WSN-BFSF that contains realistic network scenarios with various types of attacks. As shown in Fig. 2, the proposed model is performed in 3 main phases: (1) RL-

based drone-client selection, (2) local training and anomaly detection, (3) federated averaging and decision making.

5.1 RL-based drone client selection approach

Before training the proposed FDL-based IDS, we should select participating drones with desirable characteristics. So, we propose leveraging Q-learning to enable the MEC server to develop an optimal drone client selection policy within FANETs without prior knowledge of network dynamics. Q-learning is a straightforward and intuitive RL algorithm that is relatively easy to implement. Figure 3 illustrates how the MEC server (agent) continuously updates their policies through iterative interactions with drones (environment). This allows for the selection of drones based on factors such as power capacity, storage availability, and location. Our model employs an MDP framework to describe drone behavior and model the FANET environment. Then, we use Q-learning with an ϵ-greedy policy to solve the MDP problem.

5.1.1 MDP reformulation

Our model uses an MDP framework to model the FANET problem. The MDP consists of several key components: the state space ($\mathcal{S}$), the action space ($\mathcal{A}$), the state transition probability (P), and the reward function (R). In the proposed model, the MEC servers are considered as mobile agents (Non-stationary MEC Servers) responsible for observing drones and aiming to maximize the cumulative expected reward. The MDP denoted as $\mathcal{S}$, $\mathcal{A}$, P, R, can be described as follows:

– The state space ($\mathcal{S}$) includes the following components: power capacity, storage availability, and location. *Power Capacity* The drone k has a restricted rechargeable battery to store energy. Positive integers are used to represent stored energy where *Emax* is an acronym for the drone's battery's maximum capacity, and the drone battery level at the start of the time slot is shown by the variable $E_k(t)$ where

$$E_k(t) = \{0, E1, ..., E_{max}\} \tag{2}$$

Storage Availability We divide the storage availability into discrete levels, such as low, medium, and high. Positive integers are used to represent storage availability where Mmax is an acronym for the drone's memory card maximum availability, and the drone memory space at the t time slot is shown by the variable$M_k(t)$, where

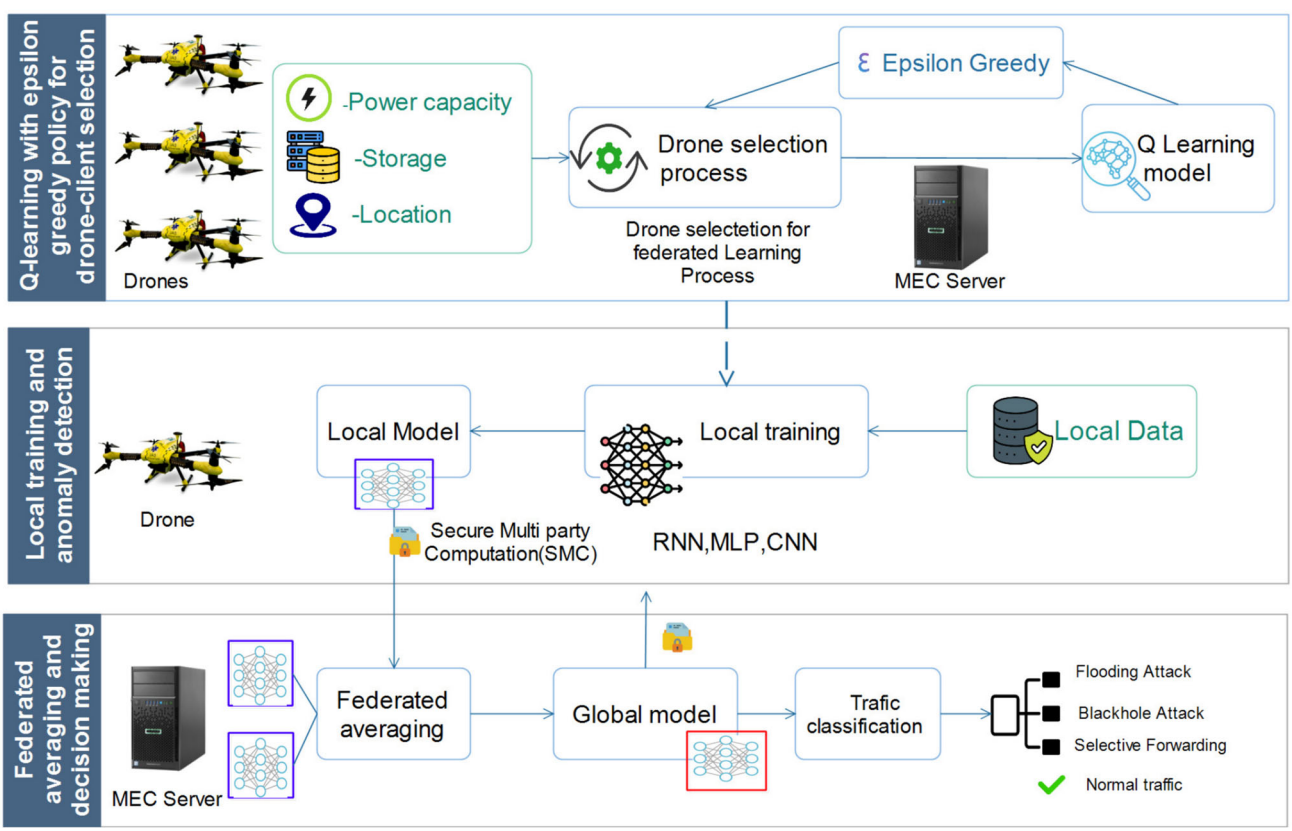


Fig. 2 FDL-based IDS (FLID) architecture

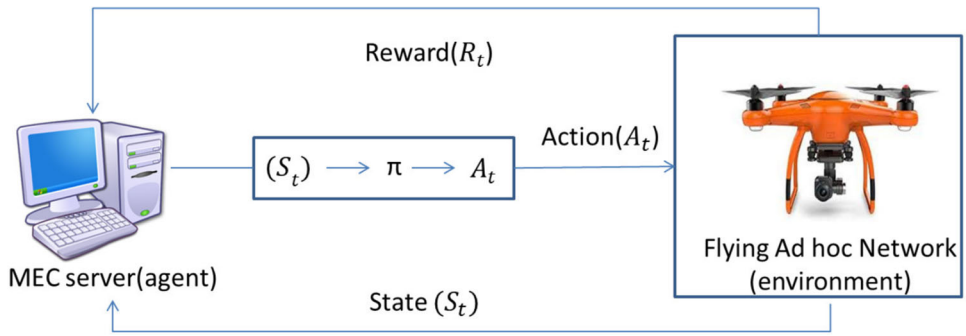


Fig. 3 RL-based drone-client selection process

$$M_k(t) = \{0, M1, ..., M_{max}\} \tag{3}$$

Location Represent the location for drone k at time slot t as discrete regions or grid cells identified by

$$\theta_k(t) = \{x_k, y_k, z_k\} \tag{4}$$

The drone's location can be associated with one of the subregions of the geographical coordinates (x, y, z). Figure 4 describes the changes in the state space of the drone as it flies from one subregion to another. Consequently, the state at any given time for a certain drone k can be a combination of the power capacity of the devices, the storage availability, and the location as follows:

$$S_k(t) = \{\theta_k(t), E_k(t), M_k(t)\}. \tag{5}$$

- Action space $\mathcal{A}$: at time slot t, each MEC server selects an action from the action space $\mathcal{A}$, according to the observed state $S_k(t)$. Let $A_t = \{a_t, a_{t+1}, ...\}$ present a sequence of actions from t to ∞. For example, action can be movement from one sub-region to another.
- The transition probability matrix P indicates the likelihood of transitioning to different states based on the drone's power capacity, storage availability, and location. The probability of a MEC server and drone moving from state s_t to state s_{t+1} during a transition step is denoted by $p(S_{t+1}|S_t, a_t)$.
- The reward R is designed to capture the selected drones' desired properties and objectives. The received reward after transitioning from state s_t to state s_{t+1} due to action a_t is defined as $R_t = \{S_t, a_t, s_{t+1}\}$. The MEC server receives a positive reward from the drone located in the same subregion, which has a high power capacity and storage availability. The MEC server receives a negative reward (penalty) from the drone located in the same subregion, which has low power capacity and storage availability. The rewards can be scaled or weighted to reflect the relative importance of each property.

We construct the associated MDP by integrating the defined state space, action space, transition probabilities, and reward function. The MDP reformulation is shown in Fig. 5.

5.1.2 Q-learning with ϵ-greedy for solving the MDP

The formulated MDP problem can be solved using the Q-learning algorithm with ϵ-greedy policy. Q-learning is an off-policy algorithm, which implies that it learns Q-values based on the maximum Q-value of the next state, regardless of the action taken in the next state. This attribute makes Q-learning suitable for tasks with exploration challenges and allows the agent to learn an optimal policy π^* even in dynamic environments. In our model, we employ Q-learning to manage large-scale decision-making as follows:

- **Step 1. Initialization of the Q-Table:** The MEC server creates a Q-table to store the action values for each state-action pair of the drone k at a specific location and time. The dimensions of the Q-table depend on the discretized state and action spaces.
- **Step 2. Initialization of hyperparameters:** In our scheme, the hyperparameters of the Q-learning algorithm are defined as follows: the learning rate (α), which determines the weight given to new information during the update of Q-values [42]; the discount factor (γ), which controls the importance of future rewards relative to immediate rewards, with $\gamma \in (0, 1)$ [43]; and the exploration rate (ϵ), which manages the trade-off between exploration and exploitation in model-free reinforcement learning [44].
- **Step 3. Exploration and exploitation:** During each episode, the agent follows the ϵ-greedy policy for action selection, where a random number between 0 and 1 is generated. If this number is less than ϵ, a random action is chosen for exploration; otherwise, the current state's action with the highest Q-value is exploited. The ϵ-Greedy algorithm provides a straightforward approach to balance exploration and exploitation by randomly selecting between them according to ϵ, with exploitation being the primary strategy and exploration occurring infrequently. This decision-making process

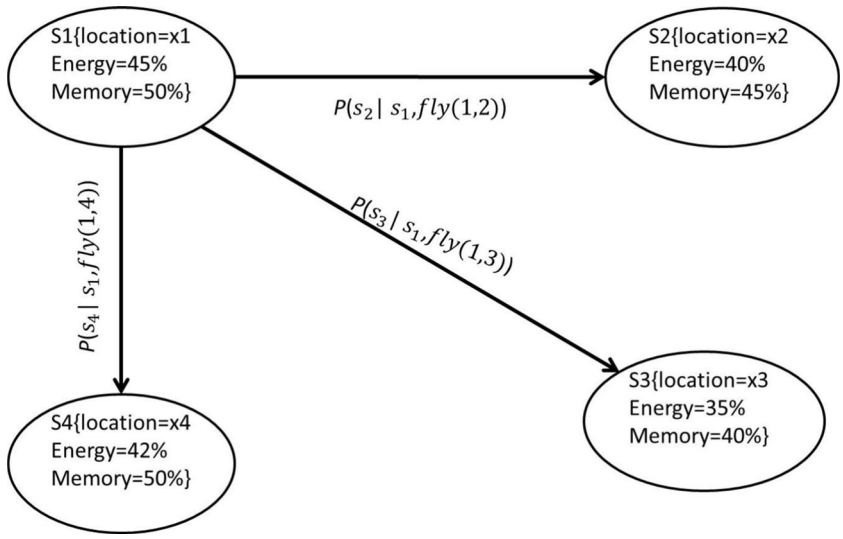


Fig. 4 State space of the drone

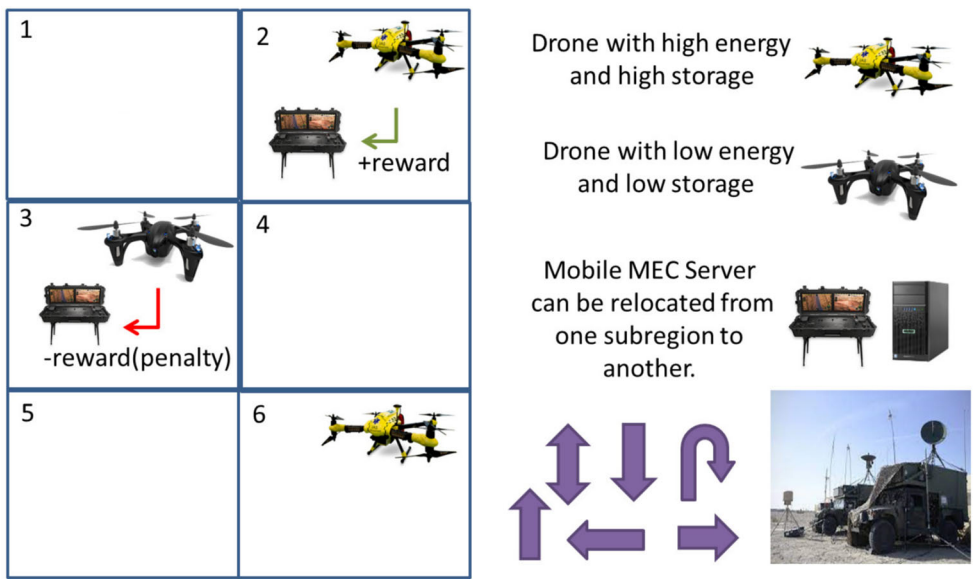


Fig. 5 The MDP reformulation process

considers both the current state and the Q-table, which stores the Q-values associated with various state-action pairs as follows:

$$\text{Action(t)} = \begin{cases} \max\limits_{a} Q_{\epsilon}(a), & \text{with probability } 1-\epsilon \\ \text{any action } (a), & \text{with probability } \epsilon \end{cases}$$

– **Step 4. Updating the Q-value of the state-action pair:** After selecting an action, the MEC server executes it within the environment, observing subsequent state changes, received rewards, and episode termination. This information is vital for updating the Q-value of the current state-action pair, which is executed using the Bellman equation [45]:

$$Q(S_t, A_t) = Q(S_t, A_t) + \alpha(R_{t+1} + \gamma max_a(Q(S_{t+1}, a) - Q(S_t, A_t))). \quad (6)$$

The parameters α and γ are selected to balance convergence speed, stability, and long-term reward optimization, ensuring effective learning while preventing excessive fluctuations. The equation in (6) combines immediate reward with maximum Q-value of the next state, discounted by a factor that accounts for the importance of future rewards.

– **Step 5. Repeat Steps 1-4:** By iteratively performing the previous steps, the agent learns and refines its Q values over time, continuously improving its decision-making process. The algorithm's convergence occurs when the Q-values stabilize, indicating that the agent has learned an optimal policy π^* for the given task.

The MEC server employs the Q-learning algorithm to select the most suitable drones to detect intrusion attacks. Subsequently, upon selecting the optimal drones, the MEC server transmits the initial parameters to initiate training of the FDL model using a split dataset. In particular, this model targets the identification of flooding, blackhole, and selective-forwarding attacks. The subsequent steps explain this procedure.

Algorithm 1 RL-Based Drone-Client Selection Algorithm

```
MEC Server:
Execute Algorithm 1
Models parameters ← values
Selected drone ← P_k, Model k
Model aggregation and execution
foreach drone-client_k do
    w_{m+1}(k) ← Drone-client(k, w)
    w_{m+1}(k) ← Σ_{k=1}^{K} (p_k / p) w^k_{m+1}
Refine the model
function Drone-client(k, w)
Local training and analysis
Split p_k into b batches of size B
foreach Local epoch do
    foreach batch b do
        w ← w − η∇(w; b)
return w
end function
```

5.2 Local training and anomaly detection

The process begins by initializing the model on the server. Initialization involves setting up the model parameters before training or learning occurs. The MEC server initializes an MLP, an RNN, and a CNN model with the appropriate architecture and hyperparameters for the traffic classification task. The MEC server sends the initial model parameters and additional encrypted instructions using Secure Multi-Party Computation to the selected drone client. These parameters are then used for drone training. Further instructions can specify the number of local training iterations that the drone should perform and the portion of local data (p_k) that should be used for training. Using its local data, the drone trains each of the three models (MLP, RNN, and CNN) separately. In the local training process, the Adam optimizer was employed due to its efficiency and adaptive learning rate capabilities. Additionally, we used the categorical cross-entropy loss function, as it is appropriate for multi-classification problems and ensures effective model convergence. The training process typically involves the following steps:

- **Step 1. Forward Pass:** The drone feeds the preprocessed traffic data through the model to make predictions.
- **Step 2. Loss Computation:** The drone calculates the loss or error between the model predictions and actual labels (e.g., normal, selective forwarding, blackhole, or flooding attacks). Each drone k trains the local model using its local data p_k and updates the model parameters based on its local training.
- **Step 3. Anomaly Detection:** The anomaly detection uses different ML models, such as MLP, RNN, and CNN. In our proposal, the data are split into B batches for each local epoch, and the drone randomly selects a subset B of the batches for training. This introduces some randomness and diversity into the training process. The drone updates the local weight w based on the training data in the selected batches as follows:

$$w_{m+1} = w_m - \eta \triangle(w_m; b), \tag{7}$$

  where $\triangle(w_m; b)$ represents the change in weight w_m for each mini-batch b, η is a hyperparameter representing the learning value that controls the size of the weight updates, and w_m is the weight at round m.
- **Step 4: Model updates** Finally, the drone sends the updated model to the MEC server using an SMC scheme to protect the privacy of its local data.

5.3 Federated averaging and decision making

In this phase, the MEC server aggregates the model updates received from the drones and creates a new global model. This updated global model is encrypted with SMC and then sent back to the drones, which use it as a starting point for their next round of local training. The drone updates w^k_{m+1} every m rounds for the drone k. A weighted average of the local updates received from the drones is calculated using the federated averaging technique. As w^k_{m+1} is received at the MEC server, these local drone updates are used to enhance the global model as:

$$w^k_{m+1} = \sum_{k=1}^{k} \frac{\mathrm{p}_k}{\mathrm{p}} w^k_{m+1}, \tag{8}$$

where w^k_{m+1} is the global weight for drone k after m rounds over all the p data points. The loss function of drone k on the local portion dataset p_k is defined as follows:

$$L_k(w) = \frac{1}{p_k} \sum_{i=1}^{k} f_i(w), \tag{9}$$

where $f_i(w) = f(x_i, y_i, w)$ is the prediction loss function from the input x_i to an expected output y_i with weight

vectors w. Accordingly, the global loss function minimization is obtained as:

$$\min_{w} L(w) = \sum_{k=1}^{K} \frac{\mathrm{p}_k}{p} l_k w, \tag{10}$$

where K is the number of participants drones in the current learning round, and $L_k(w)$ is drone k local objective function.

After the aggregation, a model trained on the combined data of all participating drones is obtained. However, this model is trained for only a short period. This process must be repeated continuously to achieve a fully trained model that performs well on all drone data. At this step, the drone can download the global weight W, representing the model parameters obtained from the aggregation. This global weight can be used during the drone's local training process. Using globally confirmed updates incorporating the aggregated knowledge of all drones, individual local drones can enhance their learning. The detailed FDL process is shown in Fig. 6.

The operation details of the proposed FDL algorithm are explained in Algorithm 2.

6 Performance evaluation

6.1 Experimental setup

We conducted our experiments on an HP Notebook equipped with an Intel Core i7 10th Gen processor and 8 GB of RAM. The primary programming language for implementation was Python, with Jupyter serving as the execution environment.

6.1.1 Simulation environment

We modeled a FANET comprising 20 mobile drones operating within a 1000-square-foot area, with a single non-stationary MEC server representing the agent responsible for drone selection. The drones operate using a random waypoint mobility model, whereby each drone chooses a random destination within the designated area and travels toward it at a steady speed. Once the drone arrives at its destination, it then selects a new random location to pursue next. The state representation of a drone consists of three key features:

– **Current location:** Represents the drone's real-time position in the simulated environment.

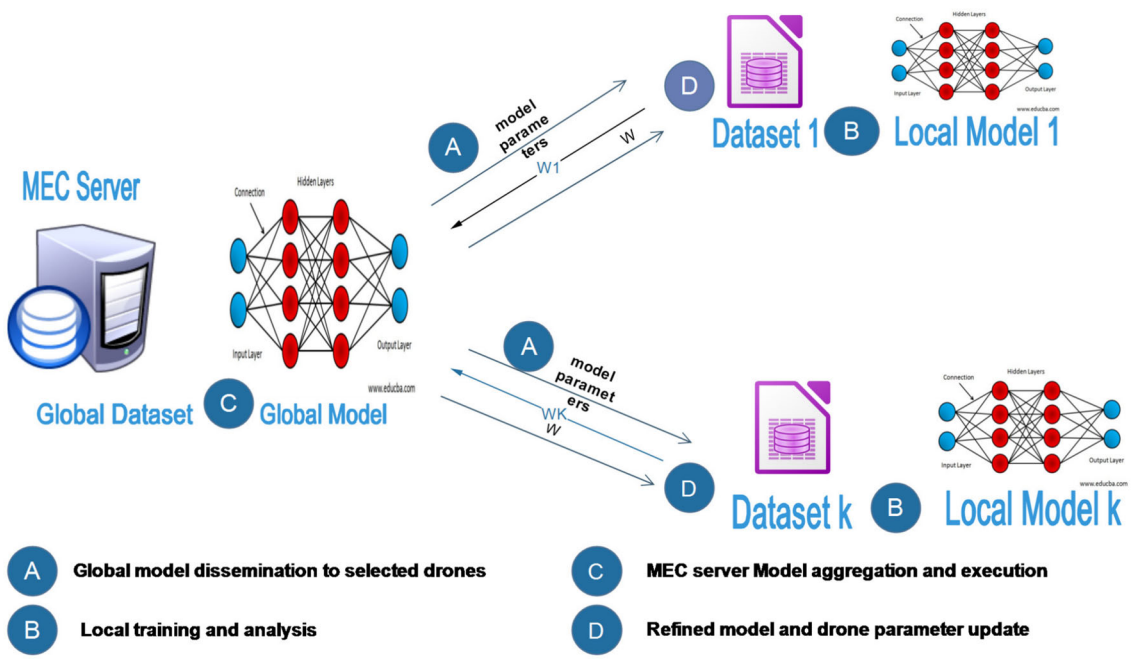


Fig. 6 FDL process

Algorithm 2 FDL Algorithm-Based IDS

```
Input: Learning rate α ← 0.5, Discount factor γ ← 0.9
Exploration vs. exploitation trade-off ϵ ← 0.1
Output: Q-table containing Q(S_t, A_t) defining the optimal policy π*
function SELECTACTION(Q, S_t, ϵ)
  n ← uniform random number between 0 and 1
  if n < ϵ then
    A ← random action from the action space
  else
    A ← arg max Q(S_t, ·)
  end if
  return A
end function
Initialization:
Initialize Q(S_t, A_t) arbitrarily except that Q(terminal, ·) ← 0
foreach episode do
  Initialize S
  foreach step of episode do
    A_t ← SELECTACTION(Q, S_t, ϵ)
    Take action A_t, then observe reward R_t and next state S_{t+1}
    Q(S_t, A_t) ← Q(S_t, A_t) + α[R_t + γ max_a Q(S_{t+1}, a) − Q(S_t, A_t)]
    S_t ← S_{t+1}
```

- **Battery level:** Reflects energy constraints, which influence task feasibility.
- **Storage capacity:** Ensures selected drones can handle the required computational load.

This problem is formulated as an RL task, where the MEC server learns a policy to optimize drone selection and maximize cumulative rewards. We employ Q-learning with an ϵ-greedy strategy to balance exploration and exploitation. The learning rate (α) is set to 0.5, and the discount factor (γ) is 0.9, chosen based on preliminary experiments indicating stable convergence within these values. We train the MEC server for 25000 episodes, with ϵ starting at 0.1 and decaying exponentially using a decay rate of 0.9998. Convergence is assessed by tracking the stabilization of Q-values.

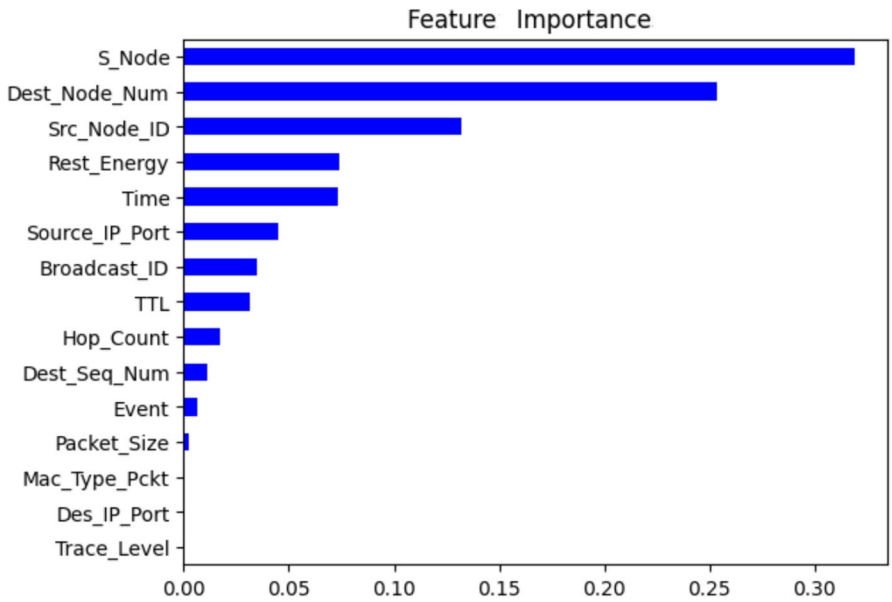
In real-world deployments, external disturbances such as interference from neighboring drones, environmental obstacles, and dynamic network conditions may affect drone selection. To address these concerns, our simulation environment incorporates stochastic noise in battery depletion rates and drone mobility patterns to model real-world uncertainty.

6.1.2 Dataset description, data-preprocessing, and simulation parameters

To our knowledge, no publicly available dataset is designed explicitly for intrusion detection in FANETs. Consequently, we used the WSN-BFSF dataset [46], introduced in 2023, which is tailored for detecting intrusions in WSNs. Although WSNs and FANETs differ in their topology and mobility patterns, they share key security vulnerabilities. The dataset includes three major routing attacks, i.e., flooding attack, blackhole attack, and selective Forwarding attack. These attacks are commonly observed in FANET environments, making the dataset a relevant choice for evaluating IDS performance. The dataset consists of 312106 instances, each with 16 distinct features. The dataset was pre-processed as follows:

1. **Handling missing values:** Missing entries were replaced using mean imputation.
2. **Categorical feature encoding:** The categorical feature 'Class' was encoded using label encoding.
3. **Normalization and scaling:** Feature values were normalized to $[0, 1]$ to prevent bias due to scale differences, and skewed data was standardized using the Standard Scaler, which is essential for deep learning methods.
4. **Class balancing:** Oversampling techniques, specifically SMOTE, were applied to address the class imbalance.
5. **Feature selection:** The correlation coefficient between each feature and the target variable was computed to measure the strength and direction of their linear relationships. Features were sorted based on their correlation coefficients, considering both positive and negative values, and the top-ranked features were selected for training. Feature importance is depicted in Fig. 7.

Fig. 7 Feature importance



6. **Partitioning and cross-validation:** The dataset was split into 80% for training and 20% for testing. Cross-validation was conducted using a 5-fold strategy to mitigate data leakage risks and improve generalization.

For training and testing the FDL model, we implemented FLID on Jupyter using:

- **NumPy** for numerical computations.
- **Pandas** for data manipulation.
- **Scikit-learn** for ML functionalities.
- **Keras and TensorFlow** for implementation of deep learning models.

The FL training process involved 20 communication rounds between drones and the MEC server. Each drone was assigned a batch size of $B = 32$ and trained locally for four epochs per round (i.e., $N = 4$). The local learning rate (LR) was set at 0.001 to ensure stable convergence. Table 2 summarizes the experimental parameters.

6.2 Impact of learning rate and discount factor on average reward

The learning rate α and the discount factor γ are critical hyperparameters in RL algorithms. These parameters influence how the agent updates knowledge and prioritizes immediate versus long-term rewards. Selecting the optimal values for these hyperparameters ensures that the algorithm converges to a high-performance policy. Figure 8(a) illustrates the relationship between α and the average reward received by an agent in a Q-learning context. The x-axis represents the learning rate, which ranges from 0.01 to 0.5. The y-axis shows the average reward, which reflects the agent's performance. The average reward increases as α increases from 0 to 0.5. The highest average reward is at the maximum value of α, (0.5), approximately -155. We conclude that the best value of α is 0.5. The algorithm achieves the highest average reward at this value, signifying optimal performance. Figure 8(b) illustrates the relationship between the discount factor (γ) and the average reward received by an agent in an RL context. The x-axis represents the discount factor, which ranges from 0.1 to 0.9. The discount factor determines how much future rewards are valued compared to immediate rewards. The y-axis shows the average reward, which reflects the agent's performance based on the chosen discount factor. For lower discount factors ($\gamma = 0.1$ to $\gamma = 0.5$), the average reward is significantly lower (around -194). The average reward increases noticeably with a discount factor of 0.7, reaching around -185. It peaks at a discount factor of 0.9, reaching approximately -181. The increase in the average reward with higher discount factors indicates that the agent can better appreciate and plan for future rewards. The peak at $\gamma = 0.9$ suggests that this is possibly the optimal discount factor for the given environment.

6.3 Performance evaluation of the Q-learning-based drone client selection

Figure 9 investigates learning performance in terms of learning progress, time complexity, and convergence. Figure 9(a) plots the negative rewards as a function of the number of episodes (learning progress). This figure consistently improves the mean reward over the 2000 episodes, indicating the agent's learning. Significant gains occur early on, with the mean reward decreasing from -178 to -134 after 4000 episodes, though it decreases later. The

peak performance is achieved with a mean reward of -34 in the final 2000 episodes after 24000 total episodes. In general, the agent shifts from exploration to exploitation, with improving rewards, demonstrating the efficacy of Q-learning in drone client selection. Figure 9(b) investigates the time complexity of the implemented algorithm. This figure plots the training time versus the training episodes. This figure shows highly variable training times, with numerous spikes and fluctuations. Some episodes significantly exceed others in training time, with the highest spike reaching around $2.5e-5$ seconds. However, most episodes have relatively short training times, often below $1e-5$ seconds. The variability and spikes likely result from the complexity of state space and the exploration process. The algorithm becomes more efficient as training progresses, leading to shorter training times for most episodes. Figure 9(c) investigates the convergence performance of the proposed learning algorithm. It illustrates the rewards received over time. Initially, performance improves rapidly, plateauing as the agent learns the optimal policy, indicating convergence. The episode rewards in Fig. 9(c) show a highly variable pattern during training, with significant fluctuations and numerous spikes near 0, along with negative rewards dropping as low as around -500. This variability is typical of Q-learning algorithms, where the MEC server explores different drone selections and receives rewards or penalties based on the outcomes, including negative rewards for suboptimal choices. The convergence to a stable point suggests that the agent has learned an effective strategy to achieve its goals. Moreover, the convergence to a stable point occurs within the context of a single-agent scenario because the environment remains static, indicating that the reward distribution for a specific action in a particular state remains constant.

Table 2 Experimental parameters

Parameter	Description
Processor	Intel Core i7 10th Gen
RAM	8 GB
Programming language	Python 3
Environment	1000-square-foot area
Training episodes	25000 episodes
Exploration vs.exploitation ϵ	0.1
Penalty	300
Reward	25
EPS decay	0.9998
Show every	2000
FDL rounds	20 rounds
Batch size (B)	32
Local epochs (N)	4
Local learning rate (LR)	0.001
Activation function	ReLU
Classification function	Softmax
Optimizer	Adam
Loss function	Categorical crossentropy

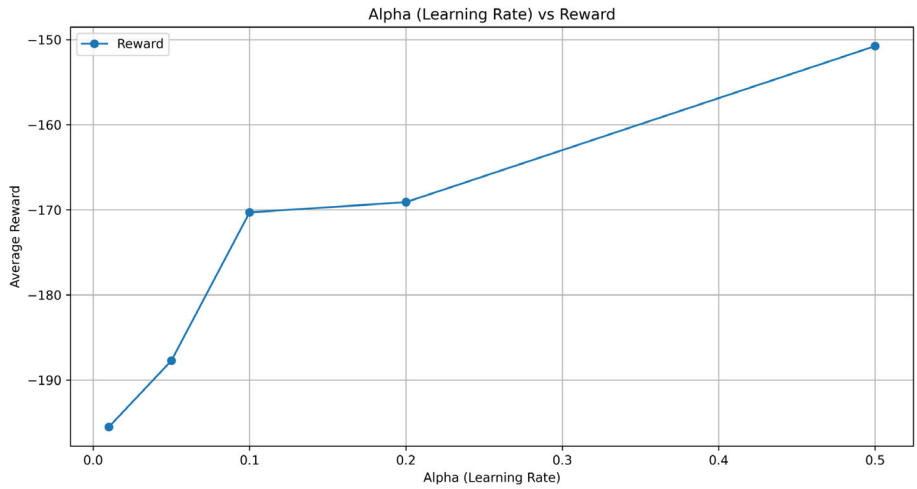
6.4 Performance evaluation of the FDL-based IDS algorithm

Table 3 provides a performance analysis of different models under FDL-based IDS. This table investigates the performance of three FDL models, RNN+LSTM, MLP, and CNN, over multiple communication rounds on a WSN-BFSF dataset. The RNN+LSTM model demonstrates the most consistent performance gains with additional communication rounds. Its precision improved steadily at each round, reaching 99% after 25 rounds. This suggests that RNN+LSTM is particularly effective at learning representations from federated distributed data in a sequential manner. Although the accuracy of the federated MLP improves more between rounds 1 and 5 than in the other models, its gains stalled after ten rounds, stabilizing near 98%. The MLP architecture may have reached the limit of what it could learn from the decentralized data more quickly. Federated CNN shows the lowest starting accuracy and indicates gradual improvements with additional rounds, leveling below 97%. The CNN's ability to discover valuable features across devices appears limited compared to RNNs and MLPs under federation.

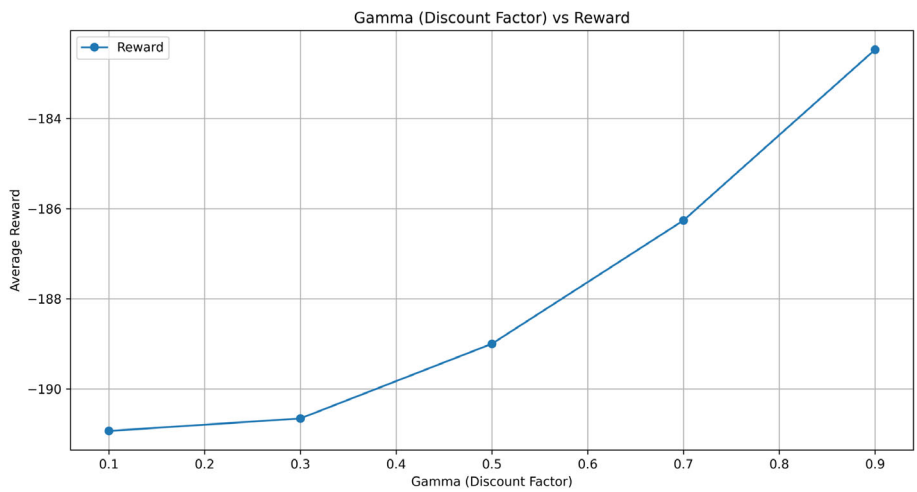
The differences in performance among the models can be attributed to their architectural characteristics. The superior performance of the RNN+LSTM model is due to its ability to capture temporal dependencies in federated data, making it particularly effective for sequential patterns. In contrast, the MLP model, despite its initial rapid accuracy gains, struggles to sustain improvements due to its fully connected structure, which lacks the temporal modeling capability of RNNs. The CNN model, which relies primarily on spatial feature extraction, demonstrates the lowest performance, as spatial relationships alone are insufficient to capture the sequential nature of network traffic patterns in intrusion detection. We conclude that the federated RNN+LSTM architecture achieves the highest performance, with an accuracy of 99.38%, benefiting from its strength in learning sequential dependencies in federated settings.

Analysis of misclassification cases We provide an analysis of the misclassification cases by investigating instances where each model incorrectly classifies data,

Fig. 8 Learning rate and discount factor vs. reward



(a) Learning rate vs. reward.



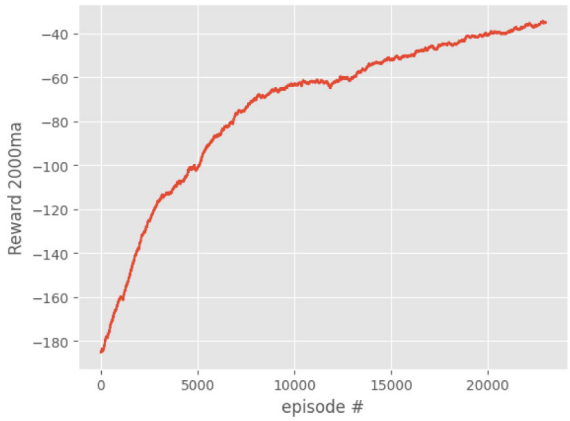
(b) Discount factor vs. reward.

aiming to understand the underlying causes and identify potential improvements for future iterations as follows:

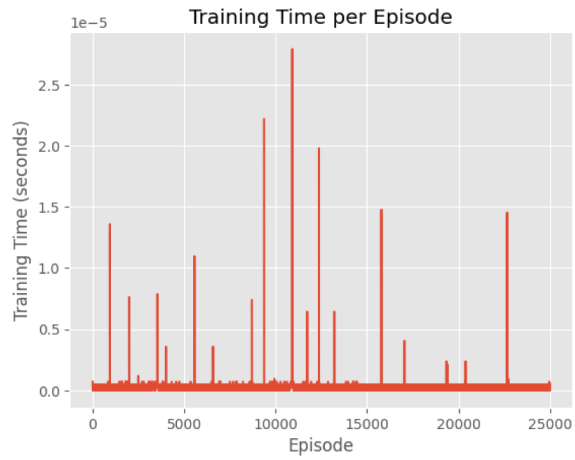
1. **RNN+LSTM:** Despite its overall superior performance (accuracy reaching 99.38%), the model occasionally misclassified packets during abrupt traffic changes or noise in sequential patterns. These cases often resulted from insufficient long-term temporal dependencies captured during training.
2. **MLP:** The federated MLP model achieved early accuracy gains but plateaued after 10 rounds. Its misclassifications occurred primarily due to its limited ability to model sequential dependencies, particularly in scenarios where dynamic traffic behaviors were present.
3. **CNN:** The CNN model exhibited the most significant misclassifications. The errors were persistent in instances that required sequential feature extraction rather than spatial relationships.

6.5 The optimal FDL model for IDS within FANETs

we evaluated our model using three types of deep neural networks: MLP, RNN+LSTM, and CNN. This integration systematically evaluates and determines the most suitable model for FANET environments. By evaluating the performance of each model under identical experimental conditions, we determine the optimal architecture in terms of precision, recall, and F1 score.



(a) Q-learning progress graph



(b) Q-learning complexity graph

(c) Q-learning convergence graph

Fig. 9 Performance evaluation of Q-learning with ϵ greedy policy

Table 3 FDL performance evaluation

Accuracy	Federated RNN+LSTM (%)	Federated MLP (%)	Federated CNN (%)
Accuracy in First Round	94.23	94.03	93.75
Accuracy after 5 rounds	95.12	95.71	94.87
Accuracy after 10 rounds	96.79	96.24	95.64
Accuracy after 15 rounds	98.23	97.06	95.79
Accuracy after 20 rounds	99.13	97.81	95.84
Accuracy after 25 rounds	99.38	98.41	96.45

6.5.1 Flooding attack

Figure 10 illustrates the percentages of precision, recall, and F1-score obtained using the three deep neural networks: MLP, RNN+LSTM, and CNN, to identify the flooding attacks. The results indicate that the MLP model achieved slightly higher precision, recall, and F1-scores, reaching 98%. The RNN+LSTM model exhibited near-perfect scores, achieving 99%. Furthermore, the CNN model also performed well, with scores exceeding 98%. These results demonstrate that the three models achieved excellent precision, recall, and F1-score performance, with the RNN+LSTM model emerging as the optimal choice.

6.5.2 Blackhole attack

Figure 11 illustrates the precision, recall, and F1-score achieved by MLP, RNN+LSTM, and CNN models in identifying black hole attacks. The MLP model achieved a precision of 88%, recall of 83%, and an F1-score of 85%. The RNN+LSTM model achieved a precision of 99%, recall of 96%, and an F1-score of 98%, significantly

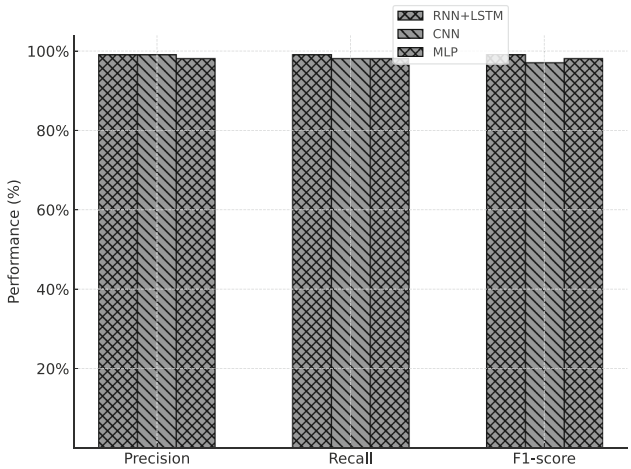


Fig. 10 Models performance for flooding attacks

outperforming the other two models. In contrast, the CNN model showed considerably lower performance with a precision of 72%, recall of 43%, and an F1-score of 54%. These results demonstrate that the RNN+LSTM model is the most effective for detecting black hole attacks, followed by the MLP model, while the CNN model performs poorly in this context.

6.5.3 Selective forwarding attack

Figure 12 illustrates the precision, recall, and F1-score percentages achieved by the deep neural networks MLP, RNN+LSTM, and CNN in identifying selective forwarding attacks. The results indicate that the MLP model attained a precision of 95%, recall of 91%, and an F1-score of 93%. In comparison, the RNN+LSTM model demonstrated slightly superior performance with precision at 97%, recall at 96%, and an F1-score of 96%. In contrast, the CNN model lagged, achieving a precision of 77%, recall of 57%, and an F1-score of 65%. The RNN+LSTM model consistently outperforms the others in all evaluated metrics, making it the most effective choice for detecting selective forwarding attacks. The MLP model follows closely as the second-best performer, while the CNN model exhibits significantly weaker performance in this context. Based on these findings, RNN+LSTM is recommended as the optimal model for detecting these types of cyber attacks.

7 Performance comparison with existing models

We first compare the computational overhead, in terms of time and energy consumption, of CDL, FDL, and FDL with client selection (FLID). Then we compare the accuracy performance metric of the CDL approach with MLP, RNN + LSTM, CNN, and their FDL counterparts. Finally, we compare the performance of the proposed FDL model with three reference models: LSTM-RNN-IDS [13], SP-IoUAV [20], and STFA-HDLID [15]. The metrics used for comparison are accuracy, precision, recall, and F1-score. These are the standard metrics used to evaluate classification models such as IDS.

7.1 CDL Vs. FDL: computational overhead analysis

We compare the computational overhead of CDL, FDL, and the proposed FLID model in terms of time and energy consumption, using the WSN-BFSF dataset and the RNN+LSTM model.

Figure 13a shows that CDL has the highest time overhead (18.1 min) due to centralized processing and communication delays. FDL reduces this to 8.57 min using distributed computation, while FLID achieves a slight improvement to 8.4 min by optimizing client selection, avoiding delays from low-performing clients.

Figure 13b highlights the energy efficiency gains. CDL consumes 741 joules due to its centralized architecture, while FDL reduces this to 347 joules by localizing data processing. FLID further improves efficiency, consuming only 173.5 joules-50% less than FDL and 76.6% less than CDL.

These results demonstrate that FDL reduces computational overhead by distributing training and transmitting only model parameters, while FLID further enhances efficiency by selectively involving clients in training rounds.

7.2 CDL Vs. FDL: accuracy analysis

Figure 14 illustrates the classification accuracy of the three types of deep neural networks (MLP, RNN + LSTM,

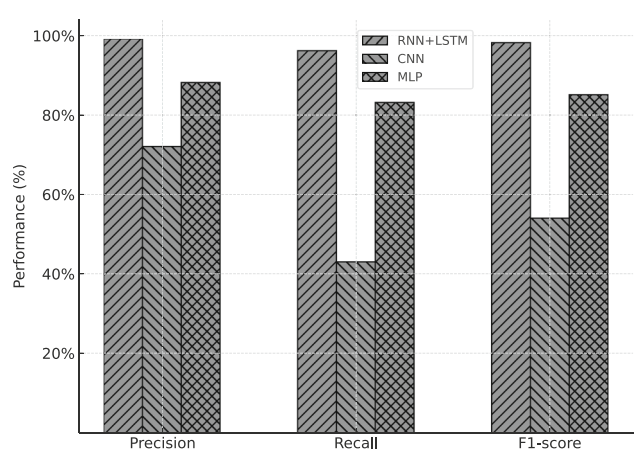


Fig. 11 Models performance for blackhole attacks

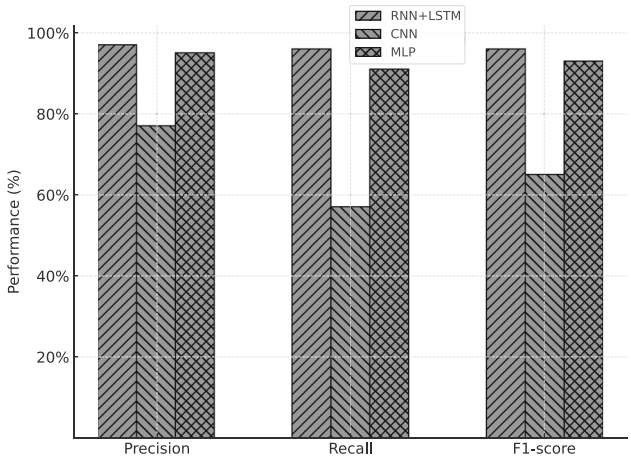


Fig. 12 Models performance for selective forwarding attacks

CNN) using CL and FDL. The FDL achieves a higher accuracy than the CDL in all three models. The RNN+LSTM architecture performs the best overall with an accuracy of 99.38% using FDL. FDL improves accuracy over CDL by 1.62 percentage points for MLP, 1.21 points for RNN+LSTM, and 0.97 points for CNN. The best performance is observed for the RNN+LSTM model. These results demonstrate that FDL is more effective than CDL for this classification. By distributing training across decentralized devices, FDL can leverage more diverse data than could be collected by a single centralized entity. The RNN+LSTM architecture provides the highest significant performance gains from FDL, suggesting that learning from distributed data sources may be particularly suited.

7.3 FLID comparison with existing models

Figure 15 compares the performance of the proposed algorithm with existing systems (SP-IoUAV, STFA-HDLID, and LSTM-RNN-IDS) in terms of accuracy, precision, recall, and F1-score. In Fig. 15a, the proposed model achieves the highest accuracy at 99.38%, followed by SP-IoUAV and STFA-HDLID, both at 99%. LSTM-RNN-IDS exhibits the lowest accuracy at 95%. The superior accuracy of the proposed model indicates its effectiveness in accurately identifying both normal traffic and attacks. Despite the high accuracy achieved by other top models, the proposed model correctly identifies all instances. Figure 15b shows that the precision of the proposed model reaches 99%, matching SP-IoUAV, while STFA-HDLID achieves 97.09%, and LSTM-RNN-IDS scores 94%. This observation highlights the effectiveness of the proposed SP-IoUAV models in minimizing false positives, i.e., incorrectly labeling normal traffic as attacks. Figure 15c reveals that both the proposed FLID and LSTM-RNN-IDS exhibit a recall of 99%, followed by SP-IoUAV with 98%, and STFA-HDLID with 97.38%. This highlights the effectiveness of the different models in identifying all attacks while minimizing false negatives, thus providing efficient attack detection. Figure 15d reveals that the proposed model achieves the highest F1-score of 99%, followed by SP-IoUAV and STFA-HDLID with a 97.37% F1-score and LSTM-RNN-IDS with 97%. The F1-score, which is a harmonic mean of precision and recall, represents a balanced performance between the two metrics. The consistently high F1-score across models emphasizes their overall effectiveness in intrusion detection. In summary, the proposed model consistently

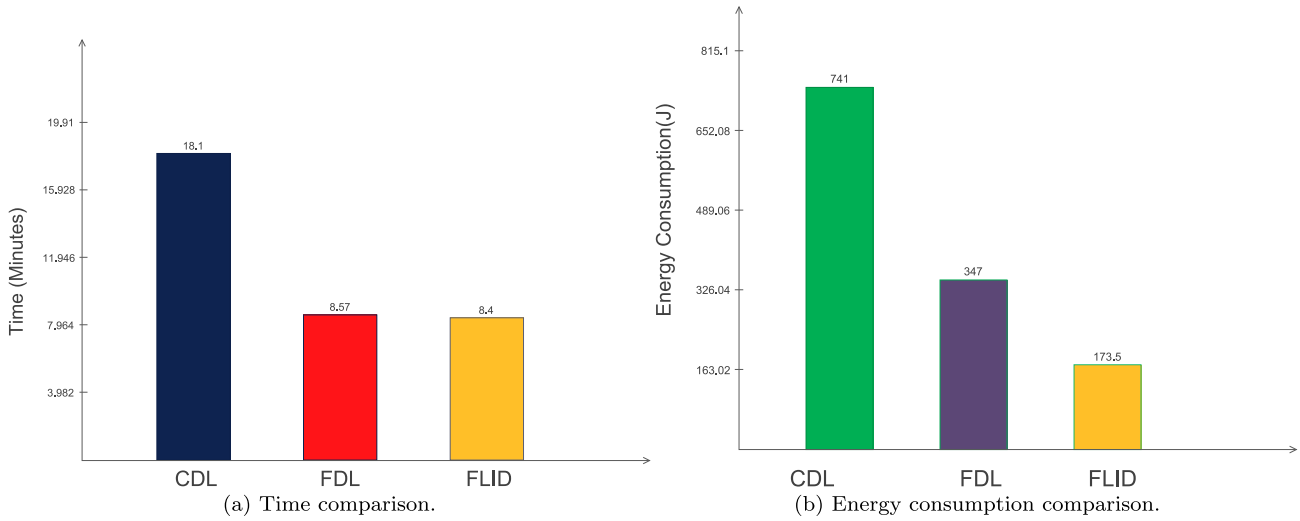


Fig. 13 Comparison of time and energy consumption

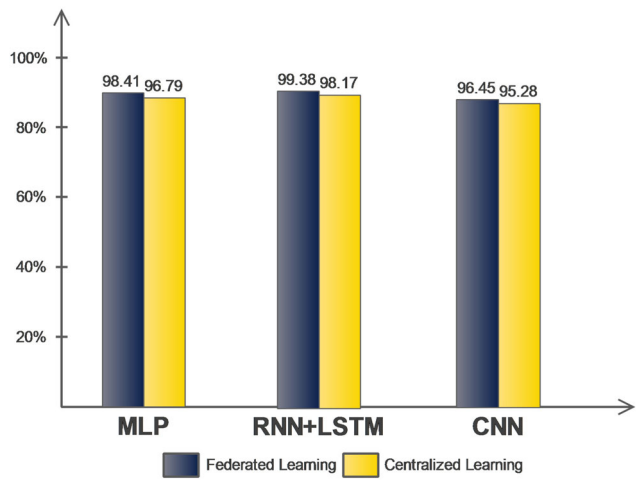


Fig. 14 Accuracy metric in CDL and FDL

outperforms other models in all metrics, achieving the highest accuracy, precision, recall, and F1-score. Although SP-IoUAV and STFA-HDLID also exhibit robust performance, LSTM-RNN-IDS shows comparatively lower performance. These findings indicate the promising potential of the proposed FLID within FANETs.

8 Conclusions and future directions

This paper investigated the design of an optimized FDL-based IDS designed for FANETs, addressing the unique challenges posed by resource constraints and communication limitations in FANET environments. Initially, a drone client selection technique utilizing Q-learning with an ϵ-greedy policy was proposed to facilitate the participation of optimal drone clients in the FDL process, effectively mitigating the aforementioned challenges inherent to FANETs. The experimental results demonstrated high convergence and low time complexity of the proposed Q-learning algorithm. Subsequently, an FDL-based IDS specifically designed for FANETs was introduced, employing three different types of deep neural networks (MLP, RNN+LSTM, and CNN). The performance of these proposed models was evaluated. The results indicated that the FDL performs notably well, particularly the RNN+LSTM model. Moreover, the proposed model meets all the essential requirements of an efficient IDS and outperforms other reference models in terms of accuracy, precision, recall, and F1-score. As part of our future research agenda, we plan to explore hybrid simulation approaches that combine real-world FANET components with simulated environments. This will enable us to evaluate the performance of our framework under more realistic conditions and ensure its effectiveness in practical deployments. Future research directions also include

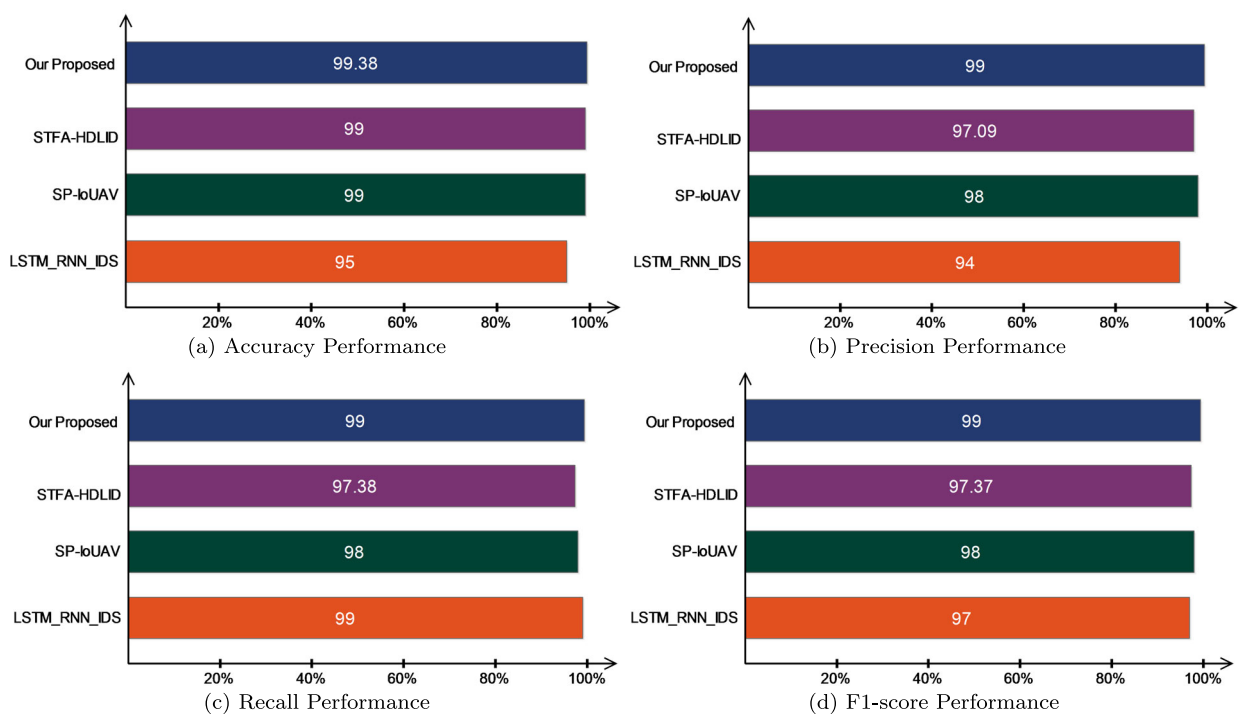


Fig. 15 Performance comparison

integrating blockchain technology into FDL to enhance IDSs within FANETs, focusing on practical implementations such as lightweight consensus mechanisms and scalable smart contracts. This integration addresses challenges such as trust management, secure data exchange, and decentralized authentication. In addition, we intend to develop dedicated datasets tailored for IDSs in FANET environments to support and advance research in this domain. These datasets will include diverse attack scenarios, mobility patterns, and communication characteristics relevant to FANETs, ensuring comprehensive model evaluation. Furthermore, we plan to incorporate robust aggregation methods, such as Krum, Trimmed Mean, or Median, to mitigate the impact of malicious updates, thus improving the resilience and reliability of the system.

Author contributions Conceptualization: [Sihem Benfriha, Nabila Labraoui, Haythem Bany Salameh, Radjaa Bensaid], Methodology: [Sihem Benfriha, Nabila Labraoui]; Formal analysis and investigation: [Sihem Benfriha, Haythem Bany Salameh]; Writing-original draft preparation: [Sihem Benfriha, Nabila Labraoui, Haythem Bany Salameh, Hassanat]; Writing-review and editing: [Sihem Benfriha, Nabila Labraoui, Haythem Bany Salameh, Radjaa Bensaid]

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Data availability No data-sets were generated or analyzed during this study

Declarations

Conflict of interest The authors have not disclosed any competing interests.

Ethical approval We state that the paper is original and was not published elsewhere.

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